

Final Project Report: TCP Modeling in Glioblastoma (CFB-GBM)

TCP Modeling GBM Project Group

Final Project Report: Tumor Control Probability Modeling in Glioblastoma

Course: TCP Modeling in GBM — Final Project

Dataset: CFB-GBM (TCIA v3, DOI 10.7937/v9pn-2f72)

Modeling cohort: n=190 (after DVH QC) · **RANO subset:** n=137

Report date: 2026-06-28 · Numbers from RESULTS.md (make report)

Abstract

We implemented a reproducible Python pipeline for Tumor Control Probability (TCP) modeling on the open CFB-GBM glioblastoma cohort. Four TCP models (Poisson, logistic, probit, gEUD) were fitted with bootstrap confidence intervals. Version 3 supplementary data provided RANO imaging response labels and author PyRadiomics features. Pooled EQD2 discriminated an OS median-split proxy (AUC0.68) but not RANO non-progressive disease (AUC0.43) when used as a dose-only predictor. After adjusting for fractionation scheme, GTV volume and radiomics features predicted early RANO with moderate cross-validated discrimination (LOOCV AUC0.64–0.74). Classical pooled TCP dose–response validation is not supported on this routine-care dataset; we provide a feasibility checklist for future open-cohort TCP studies.

Keywords: glioblastoma; TCP; DVH; RANO; CFB-GBM; open data

1. Introduction and Clinical Background

Glioblastoma (GBM) is the most common malignant primary brain tumour in adults. Standard treatment includes maximal safe resection, radiotherapy, and concurrent temozolomide (Stupp protocol: 60 Gy in 30 fractions) [1]. In elderly or poor-performance patients, hypofractionated schedules (e.g. 40 Gy in 15 fractions) are commonly used [2,3,15].

Tumor Control Probability (TCP) models estimate the probability of sterilising clonogenic tumour cells as a function of dose and radiobiological parameters [4,5]. Valid TCP studies require (i) a well-defined tumour control endpoint, (ii) inter-patient dose variation, and (iii) adequate sample size [7].

The CFB-GBM dataset (Centre François Baclesse, n=264) provides MRI, RTDOSE, GTV contours, clinical covariates, and RANO response labels [8–10]. **Project aim:** build a standard TCP pipeline

on this open cohort and assess whether classical dose–response validation is feasible; if not, identify where prognostic signal remains.

2. Dataset Description

	Item	Value
Source	TCIA CFB-GBM	Version 3
Total patients (TSV)		264
Included for modeling		190
Excluded		74 (unknown dose, missing imaging, etc.)
DVH QC exclusion		1 (patient 32, Dmean=0 Gy)
Fractionation		120 × 60 Gy/30 fr; 61 × 40.05 Gy/15 fr
Median age		70 years
Median OS		51 weeks
RANO t0→t1 labels		137 / 190

Imaging format: Pre-processed **NIfTI** (skull-stripped, co-registered), not raw DICOM-RT. RT-PLAN/RTSTRUCT DICOM are not in the public package. **GTV-only** contours at t0; CTV/PTV not provided.

3. Data Acquisition and Curation

1. Downloaded clinical/treatment/RANO/PyRadiomics TSVs (`python -m src.data.download_clinical_data`).
2. Built `cohort.csv` with inclusion rules (`cohort_builder.py`).
3. Downloaded RTDOSE + GTV t0 NIfTI for 191 patients (IBM Aspera).
4. Verified completeness (`verify_raw_data.py`); extracted DVH metrics (`feature_builder.py`).
5. Merged into `modeling_table.csv` (190 × 58 columns).

EQD2 computed with $\alpha = 10$ Gy for GTV [6,11].

4. Dosimetric Feature Extraction

From registered RTDOSE and t0 GTV masks we computed:

- Cumulative **DVH** and scalars: D95, D98, D50, D2, Dmean, Dmax, GTV **volume_cc**, Vx, **gEUD** ($\alpha = 10, 1, 10$), homogeneity index (HI).
- **EQD2** for cross-scheme comparison.

Within the 60 Gy arm, GTV Dmean SD = **0.28 Gy** — essentially no dose escalation within protocol.

5. Mathematical Description of TCP Models

Four binary-outcome TCP formulations [4–7]:

- **Poisson TCP:** clonogenic cell survival; parameters D50, 50.
- **Logistic TCP:** sigmoid dose–response; D50, k.
- **Probit TCP:** normal CDF link; D50, .
- **gEUD TCP:** dose metric from full DVH, then Poisson-style TCP; extra parameter a.

Full equations: `reports/manuscript_equations_fragment.tex`.

Estimation: maximum likelihood; **95% CI:** bootstrap (1000 resamples, seed 42).

6. Parameter Estimation and Model Comparison

	Model	AIC	ROC	AUC	Brier	Notes
Probit TCP	241.73		0.684	0.218		Best AIC (marginal)
Logistic TCP	241.75		0.684	0.218		Poisson
Poisson TCP	241.82		0.684	0.218		D5053 Gy, 503.3
gEUD TCP	243.69		0.685	0.218		3 parameters

Bootstrap Poisson (EQD2, OS proxy): **D50 = 53.2 Gy** [49.5–56.7]; **50 = 3.32** [2.06–4.69].

7. Results

7.1 Clinical prognosis

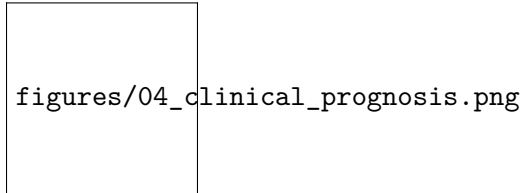


Figure 1: Figure 1

Figure 1. Kaplan–Meier OS by fractionation scheme and WHO performance status. Sixty Gy patients had median OS 60 weeks vs 28 weeks for 40 Gy (log-rank $p < 10^{-3}$). Cox model: scheme HR0.54 ($p < 0.0007$), WHO PS HR1.42 ($p < 0.001$).

7.2 Pooled TCP — OS proxy

Figure 2. TCP dose–response curves for EQD2 vs OS median (51 weeks, $n=190$). AUC0.68; LR $p < 10^{-3}$. Interpretation confounded by fractionation ($r(\text{age, EQD2})=0.57$).

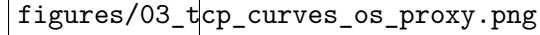


Figure 2: Figure 2

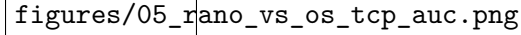


Figure 3: Figure 3

7.3 Pooled TCP — RANO endpoint

Figure 3. Same Poisson TCP on 137 RANO-labelled patients: OS proxy AUC0.62; **RANO non-PD AUC0.43** (no signal). Higher EQD2 associated with more early PD at t1 (27% vs 15% by scheme).

7.4 Within-arm analysis

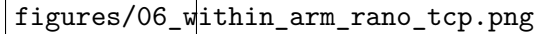


Figure 4: Figure 4

Figure 4. Stratified by fractionation: dose metrics flat within arm; **GTV volume** predicts RANO in 40 Gy arm (n=34; Poisson AUC=0.83, p0.037).

7.5 Multivariable and pooled RANO models

Figure 5. 40 Gy arm multivariable logistic ROC (volume + age + PS; in-sample AUC=0.90).

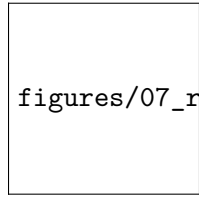
Figure 6. DVH volume validation vs author RANO volumes (=1.00, n=141).

Figure 7. Pooled models (n=137): volume + clinical + scheme AUC0.72; LOOCV0.64.

7.6 PyRadiomics comparison

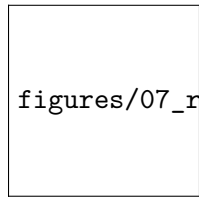
Figure 8. Author PyRadiomics (t1gd GTV t0) top-5 features vs DVH volume (in-sample AUC 0.78 vs 0.71).

Figure 9. Nested 5-fold CV reduces optimism: radiomics nested AUC0.74 vs volume 0.70.



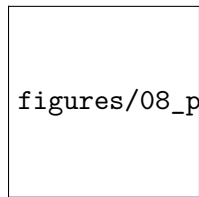
figures/07_rano_logistic_roc_40gy.png

Figure 5: Figure 5



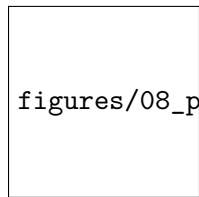
figures/07_rano_volume_validation_40gy.png

Figure 6: Figure 6



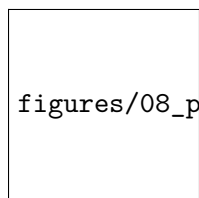
figures/08_pooled_rano_roc.png

Figure 7: Figure 7



figures/08_pyradiomics_vs_volume_auc.png

Figure 8: Figure 8



figures/08_pyradiomics_nested_cv_auc.png

Figure 9: Figure 9

8. Survival Analysis

Kaplan–Meier and Cox proportional hazards models (lifelines) included EQD2, Dmean, age, sex, WHO PS, fractionation scheme, and RANO non-PD.

Covariate (Cox, n=137)	HR	p
RANO non-PD	0.48	0.0009
WHO PS	1.57	0.0005
EQD2	0.97	0.009

RANO non-PD predicts OS independently of dose and performance status.

9. Literature Review and TCP Parameter Comparison (Part VI)

Literature table: `reports/literature_table.csv` (18 references). DOI verification: `reports/literature_doi_ch`

Source	D50 (Gy)	50	Endpoint	Comparable?
This study (CFB-GBM)	53.2	3.32	OS median (proxy)	reference
Maitre et al. 2020 (review)	40–80	1–5	LC / NTCP	partial
Okunieff et al. 1995 (meta)	~50–70	~1–4	Tumour control	partial
Ohri et al. 2017	—	—	LC preferred	design note

Our D50 falls in published ranges but is **not interpretable as GBM tumour-control D50** without local control endpoint and single-protocol dose spread.

10. Discussion

Why pooled dose–TCP fails: CFB-GBM reflects routine practice, not dose escalation. Within-arm GTV Dmean variation <1 Gy. OS and early RANO capture different biology; fractionation is confounded with age and treatment intent [7,15].

Where signal remains: GTV volume and PyRadiomics predict RANO when scheme is adjusted (pooled LOOCV AUC0.64; 40 Gy LOOCV0.74; PyRadiomics nested CV0.74). This reframes the project from classical TCP to **tumour burden → imaging response** — clinically interpretable and aligned with Moreau et al. radiomics work on the same dataset [9,10].

Limitations: No formal local control; single institution; small 40 Gy RANO subset (n=34); t1 GTV NIFTI not validated locally.

Feasibility checklist for future open TCP studies [7]: report within-arm Dmean IQR; prefer LC or RANO over OS; adjust for fractionation; report LOOCV/nested CV; compare against author radiomics baselines.

11. Conclusion

We delivered an open, reproducible TCP modeling pipeline on CFB-GBM and demonstrated that **classical pooled TCP validation is not feasible** on this cohort. With confounding addressed, **GTV volume and PyRadiomics features predict early RANO** with moderate cross-validated discrimination. All code, figures, and auto-generated metrics regenerate via **make report**.

References

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Appendix — Reproducibility

```
pip install -r requirements.txt
make report                                # regenerate RESULTS.md
python scripts/verify_literature_dois.py  # check reference links
bash scripts/run_notebooks_check.sh       # execute notebooks 01, 03-06
bash scripts/export_assignment_report.sh  # this document → DOCX/PDF
bash scripts/export_manuscript.sh         # scientific manuscript export
```

Group glossary (plain language): group_glossary_guide.md